

Virtual Karaoke Project Report



***Prepared by
Michael Ahmadi
Jacob Gepp
Bhumit Patel
Jayanth Podapati
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University of Illinois Chicago***

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I Project Description

1 Project Overview

Virtual Karaoke is a project that allows users to easily engage in live karaoke with other users over the internet. The users will be allowed to create or join live rooms, wherein the karaoke will take place. Inside the live rooms, the users can take advantage of functionality such as queues for singing, an integrated method of playing songs, and microphone/camera input transmission in order to take turns singing their favorite songs. The application looks to emulate in-person karaoke closely and in a streamlined and easy to use way.

2 The Purpose of the Project

2a The User Business or Background of the Project Effort

Karaoke is an entertainment business practiced and recognized around the world. There are two ways a karaoke business can operate. In one case, a karaoke business can let out private rooms to separate groups of customers. Each room is equipped for karaoke, meaning comfortable furniture, a large TV or projection screen to put song lyrics on, speakers, and microphones are provided. Each group pays for a room and can allow peruse a menu to purchase refreshments while inside. The second case in which a karaoke business can exist is a karaoke bar. The main difference between the two is that a karaoke bar offers a public singing experience instead of a private one: instead of singing in private rooms to their friends, people sing in front of the whole bar.

Due to the Covid-19 pandemic, in-person karaoke is no longer viable from a health perspective to many people. Karaoke businesses may also be forced to stay closed due to pandemic restrictions put into place by local governments. People turning to online karaoke have no choice but to repurpose online meeting software for a karaoke experience, which is often unideal. This application hopes to emulate the in-person karaoke experience that is currently unviable and provide an optimal experience for those seeking an online solution.

2b Goals of the Project

The client needs to adapt to the online-oriented state of our current society. Even after the Covid-19 pandemic is over, it is likely many aspects of society that were transferred online will continue to stay at least partially online. The goal of this project is to fill the market gap for online karaoke and strive to offer the best online karaoke experience so that the application remains successful even after the Covid-19 pandemic ends.

2c Measurement

The measurement of the success of this project will be the revenue generated from the project. The application can be monetized in several ways: it could be sold as a one-

time purchase, or as a subscription service, or advertising space could be sold to advertisers. A second measurement of success will be the number of daily users of the application, and a third measurement of success will be user reviews. If the project generates a favorable amount of revenue, or if it has many daily users and positive user reviews, then the project can be considered successful.

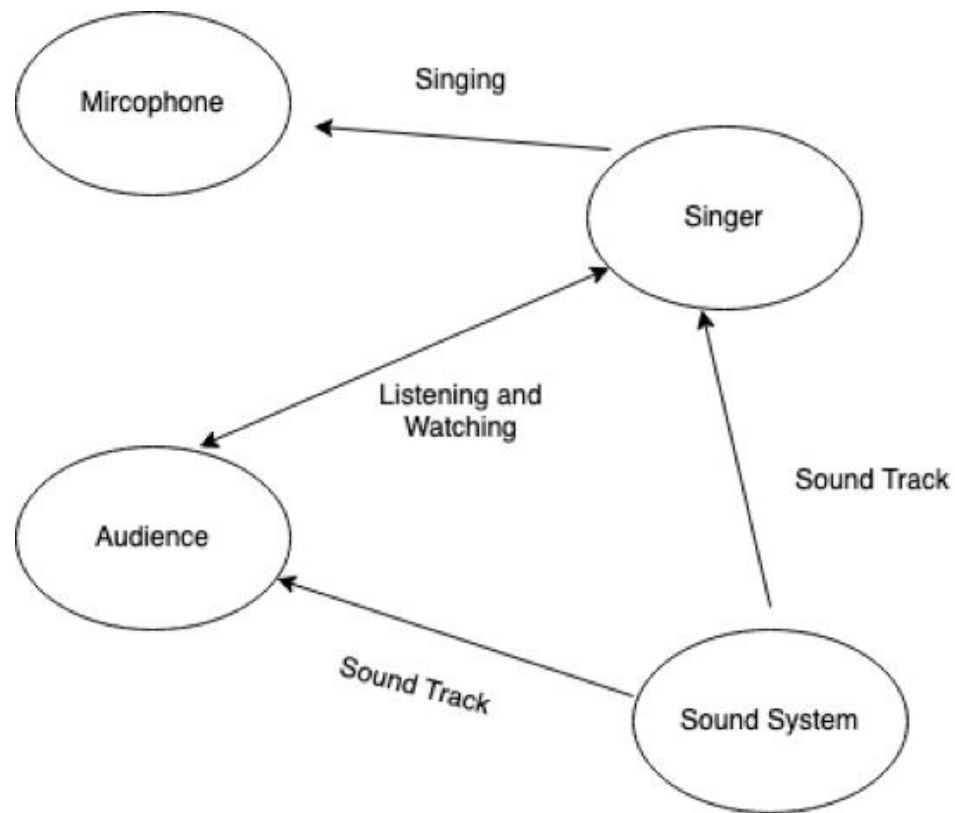
3 The Scope of the Work

Karaoke is an interactive entertainment where a person sings along with a soundtrack. Interactive entertainment has transformed significantly with the growth of technologies allowing for less physical interactions and creating more addictive forms of entertainment activities. Although concerts are still widely popular, social media has gained a monopoly on interactive entertainment with its ability to quickly provide new content at any time and anywhere. Interactive entertainment has evolved to be done just with a click of a button.

3a The Current Situation

Karaoke is usually done in public places such as bars and restaurants. People would do Karaoke in public places, however due to COVID restrictions this is not possible anymore. With the growth of applications that allow people to order food from a nearby restaurant in seconds, people are not as likely to eat at restaurants. With social media to keep people entertained, the popularity of socially engaging events like Karaoke has gone down. In order to address these issues, our project allows users to do Karaoke online giving people the convenience to do Karaoke from their laptop, phone, or tablet from anywhere at any time.

3b The Context of the Work



The Diagram contains four important components: the microphone, singer, audience, and sound system. These components are essential in the “work” of karaoke. The diagram models how the components interact.

3c Work Partitioning

Event Name	Input Output and	Summary
Watching the singer	Visual (in/out)	Audience should be able to watch the singer.
Watching the audience	Visual(in/out)	Singer should be able to watch the audience.
Listening to the singer	Audio(out)	Audience should be able to listen to the singer.

Playing Soundtrack	Audio(out) Input to start the soundtrack(in)	Audience and the singer both should be able to hear the soundtrack.
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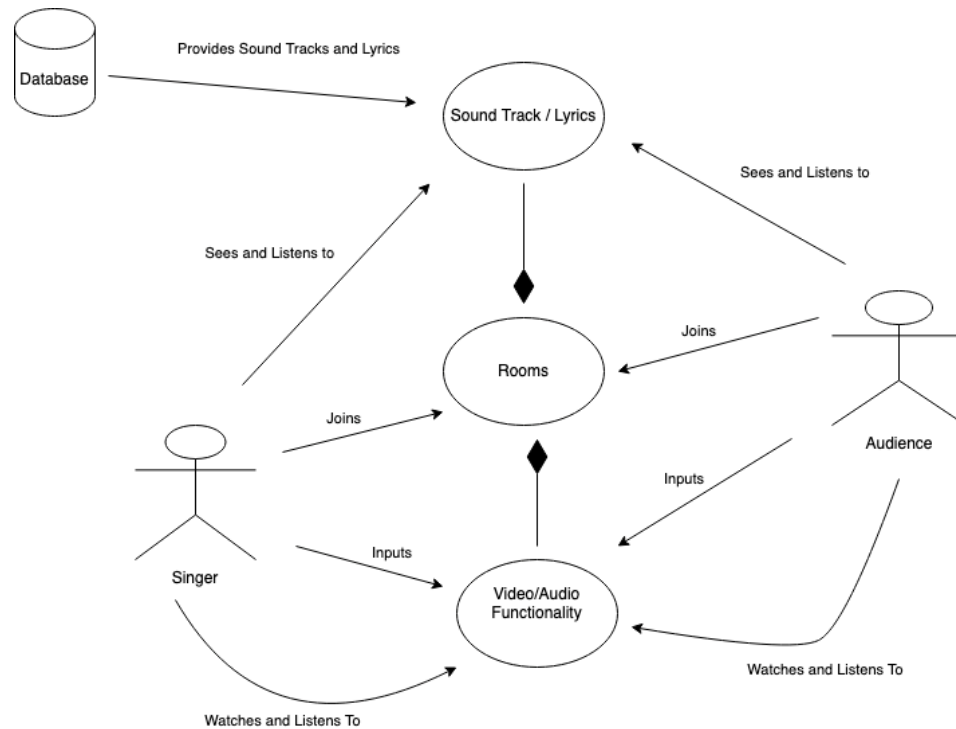
3d Competing Products

The competing products to our product are those that allow people to communicate through video and audio online. Products like Zoom, Microsoft Teams, or Discord are competing products where it is possible to do Karaoke virtually. However, these products do not stream video at very high quality and the sound quality is also not optimal. These products fail to create an environment which is suitable for karaoke due to poor video and sound quality. Our products will ensure high sound quality by having a database containing high quality soundtracks. Our fast servers will deliver video content to users without depreciating video quality. Our platform's high video and sound quality will ensure users choose our platform for Karaoke over other platforms that allow for digital communication.

4 The Scope of the Product

Our product transitions Karaoke from a physical environment to digital environment by allowing users to get together online and do Karaoke. Users will be able to see and listen to each other through the microphone and the camera on their device. Our products ensure that the users will be able to enjoy a similar experience doing karaoke that they enjoyed physically. Our product will allow users to sign up, join rooms, listen, see, and talk to other people in rooms, play sound track and start singing.

4a Scenario Diagram(s)



4b Product Scenario List

1. Rooms
2. SoundTrack and Lyrics
3. Video and Audio Functionality

4c Individual Product Scenarios

Rooms

Users should be able to join rooms. Rooms are the environment where users will be able to listen and see each other, start the sound track, and see the lyrics. Each room will have a host that will be in charge of the room, usually the host will be the singer. By default, only the hosts microphone will be turned on, other users will be turned off. A room can be either private or public, if one wants to join a private room they will either need to be invited or room id and a password. The rooms will be designed to suit the Karaoke environment, and they are an essential component to this product.

Video and Audio Functionality

After users join the rooms, they should be able to see and listen to each other. This functionality will use the input from the camera and microphone of a user's device and send it to other users that are in the same room. Since our product aims to reduce the gap in quality between physical karaoke and virtual karaoke, a high quality camera and

microphone will be needed. Our servers will be programmed to deliver and process data rapidly to reduce lag, and make sure the user gets a consistent stream in steady flow. The video stream of the singer will be displayed on a bigger frame than other users to create a suitable visual environment. Additional features such as autotune can also be added later to increase the sound quality.

SoundTrack and Lyrics

When the singer clicks the start button, the soundtrack should start playing. The soundtrack will be heard by all the users. Along with the soundtrack the corresponding lyrics will be displayed on the screen. The soundtracks can be chosen from the database by the user. The database will contain soundtracks of high quality that will be maintained by a sound engineering team. Lyrics will be displayed in a way that they will change as the soundtrack progresses. To make the environment more appealing, we can even add visuals while displaying lyrics.

5 Stakeholders

5a The Client

The ideal goal for this product would be to reach as many people as possible, and not restrict users based on their platform. As such, the ideal client to be able to back this high ambition would be Google or Microsoft. Both have a multitude of successful musically themed apps published and could ensure this product reaches as many people as possible.

5b The Customer

The customer is anyone who has a mobile device and also enjoys singing/music with their friends and family. People who frequent karaoke bars are likely to enjoy this product.

5c Hands-On Users of the Product

- User Category: Karaoke and singing enthusiasts
- User Role: The user will navigate through the products UI, select a song, and sing along with the prompted lyrics.
- Subject matter experience: The user will not be required to have any prior knowledge of singing or of the lyrics. They will be expected to be a novice but will not be at a disadvantage if experienced.
- Technological experience: The user must be familiar with downloading applications on their mobile devices. Older generations may have difficulty doing so, but typically is common knowledge among the other age groups.
- Other user characteristics: As described above, older users will likely have a harder time acquiring the product, but such people are not likely to have a compatible mobile device. Age requirements will be placed on explicit songs and will limit content to younger users. Those who are shy about singing (regardless of enjoyment) may feel averse to this product as well.

5d Maintenance Users and Service Technicians

Service technicians will be mainly responsible for updating the list of songs with lyrics. Pop culture moves fast, so the volume of songs being added could potentially be high. This service technician would be required to have knowledge of how to add songs to the systems library. They would not be required to understand how the overall system works.

5e Other Stakeholders

Stakeholder: Google/Microsoft

As the product aims to be available on their mobile devices through their software, approval will be required to publish this product as an app. This means following the guidelines set by Google/Microsoft and developing around such guidelines. Guidelines and requirements may vary depending on the platform and will need to adapt accordingly.

5f User Participation

As the user requirements are very low, the user is not expected to spend much time to meet the requirements. The knowledge some might need to acquire is to understand how to download an app on their mobile device, and then to be able to open said app. This is not a lengthy process and could be explained in a few short steps to users who are not familiar with phone apps.

5g Priorities Assigned to Users

- Key Users: Those who enjoy singing, specifically those who frequent karaoke bars, will be the most important demographic. Such users will be the typical product consumers, and will spend the most time with the product.
- Secondary Users: Users that frequent their mobile device's app store may stumble across this product. This user may or may not be enthusiastic about singing, but may try the product out of curiosity.
- Unimportant Users: Those who do not have a compatible mobile device and/or those who do not enjoy singing will not use this product.

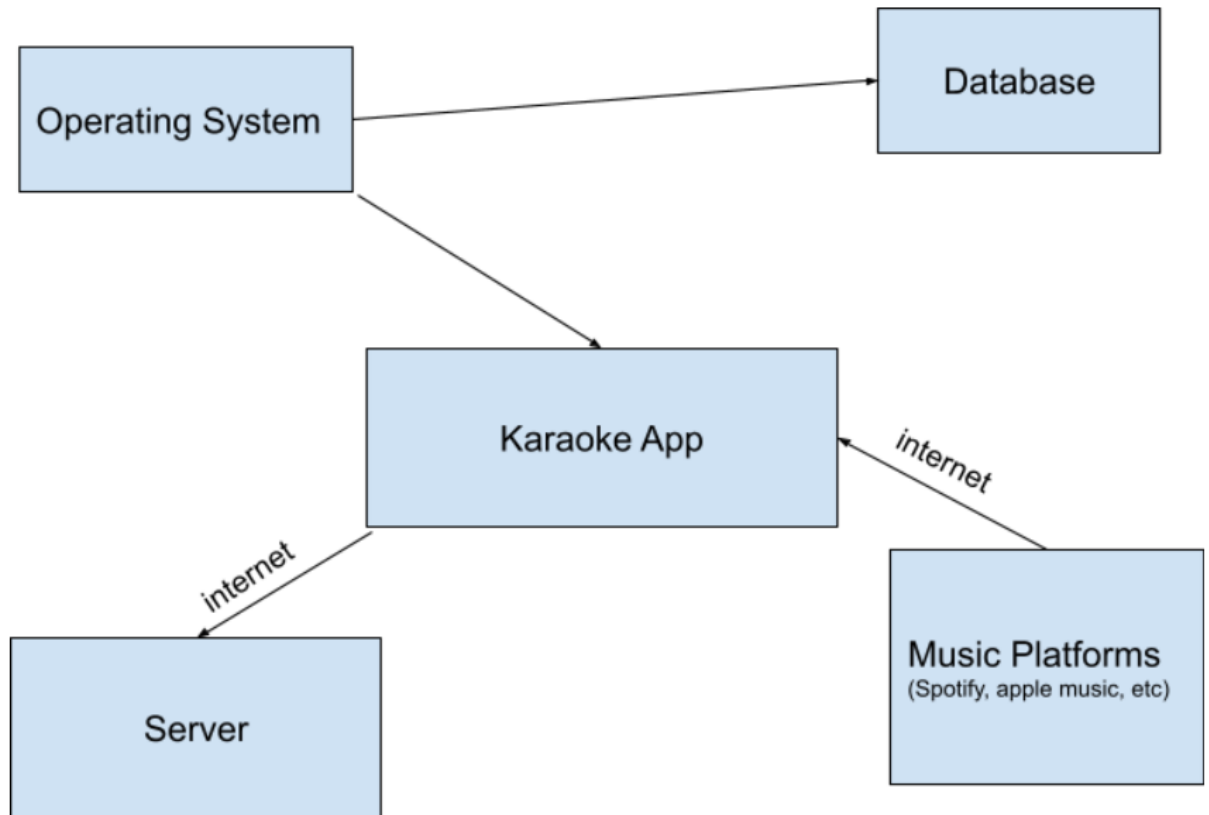
6 Mandated Constraints

6a Solution Constraints

- Basic system must be usable by any laptop that can access the internet
 - R: User is not going to buy a new device solely for this product
 - FC: Visuals and audio coming from the app should be understandable and of the best quality that the laptop can provide

- The product should be able to connect to spotify, youtube, sound cloud, apple music, etc
 - R: Most people already have a whole bunch of favorite songs. Will not appreciate having to go and input every song one by one. Very tedious
 - FC: The product should have approval from all the streaming channels.
- The product should be able to be used through any kind of smartphone
 - R: Using a smartphone is much simpler and faster than using a TV/laptop for events like karaoke where going through a whole list of songs, adding to the queue, removing from the queue, etc is involved.
 - FC: App should be designed so that a connecting app is approved and released on playstore, apple store, etc
- Should be able to connect with other accounts
 - R: Karaoke is something that is usually enjoyed with other people. The main purpose of this app is to connect one with their friends and family.
 - FC: Should be secure. The other side should be able to hear the singing properly without disturbance.

6b Implementation Environment of the Current System



6c Partner or Collaborative Applications

Music Platforms:- The karaoke project we are trying to implement should be able to access personalized playlists on music platforms like Spotify, Apple Music, etc. It should be able to read through our playlists and produce an identical playlist on the app. This can be achieved by communicating with the respective companies of the music platforms.

Webcams:- An added feature of this product is capturing the singer and showing the singer on screen. It will also allow the feature to morph the background to something the user wishes. An example would be if the user wants to see himself singing on stage, he could morph the background and it would appear as if he was standing on a stage when in actuality he is in his living room. This can be acquired by partnering up with another company specializing in webcam programming.

Operating Systems:- In order to use the app more efficiently, there should be a feature which lets us use our smartphones as a remote for the app. Flipping through the

numerous songs, choosing the background, who to talk to, etc. To do this, the app needs to be able to interact with the operating systems of different smartphones like Apple OS, Android OS, etc. To do this, we must create apps utilizing their OS and put it in the respective app store (or playstore if it is Android).

Servers and Databases: The product must store a multitude of songs and lyrics inside its system. Not to mention it should store a user's information, a round information like score, etc. And servers can be used for players to interact with each other.

6d Off-the-Shelf Software

MySQL: With one of the world's most popular databases, MySQL would definitely be an off-the-shelf software that must be included in order for the product to function properly. MySQL would let developers easily query through the data and store data into the database for the application to access.

ManyCam: An existing software webcam which allows the user to blur or replace the background without the use of a green screen. An invaluable software that must be included as it would allow users to morph their background to whatever they want.

6e Anticipated Workplace Environment

The product should be able to be enjoyed by an entire family and be used in a family setting. The output of the product should be loud enough that it can be heard through the bustling of entire families or groups of friends. While it can technically be used inside or outside, it is designed for inside use.

The workplace of the product might be noisy as karaoke is meant to be done with multiple people, therefore visual signals must be employed rather than audio signals. An example would be displaying the next person in line to sing on the screen.

6f Schedule Constraints

There will be a fixed release date for the product that the developers need to meet. By this date, the product should be fully functional and successfully finished testing. In order to meet this deadline, there will be scheduled check-in dates periodically where developers must finish the required portion by the scheduled dates.

6g Budget Constraints

There will be a limited budget in which the entire team will try their best to successfully complete the project. The budget will be split into creating successful servers for users to interact without being in person, creating the software for the project, and negotiating with the other companies.

7 Naming Conventions and Definitions

7a Definitions of Key Terms

Karaoke:- A type of interactive entertainment where people sing along to songs. A machine usually provides instrumental music along with lyrics.

Servers:- A device that provides a functionality for other programs and devices.

Database:- An organized collection of data stored electronically.

Instrumental:- The background music accompanied with a song without the lyrics without.

7b UML and Other Notation Used in This Document

This document generally follows the UML format described in “Object-Oriented Software Engineering” by Bruegge & Dutoit.

7c Data Dictionary for Any Included Models

Database - Contains user information for the server including their preferences, playlists, songs that they have sung, friends, etc.

8 Relevant Facts and Assumptions

8a Facts

- Covid restrictions are in place and CDC requirements must be met for development.
- There are over 97 million known songs and counting.
- Spotify has over 50 million songs, and adds about 40,000 new ones per day

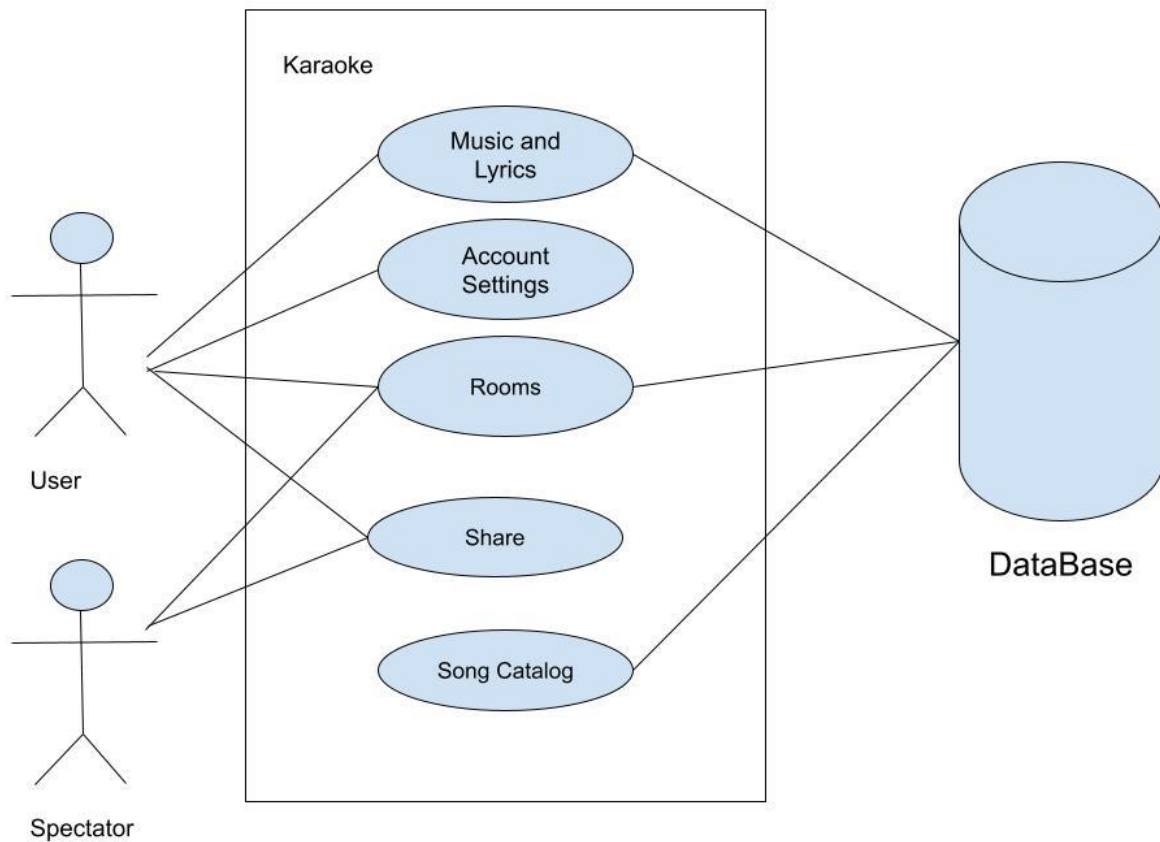
8b Assumptions

- Most users will want popular songs and genres over lesser-known music.
- Mobile devices are expected to have a working microphone and front camera.
- Webcams (with microphones) are widely available and inexpensive to the user.

II Requirements

9 Product Use Cases

9a Use Case Diagrams



9b Product Use Case List

1. Music and Lyrics
2. Account Settings
3. Karaoke Room
4. Save and Sharing
5. Song Catalog

9c Individual Product Use Cases

Use case ID: 1
 Name: Music and Lyrics
 Initiated by: User
 Triggering Event: User Selections
 Additional Actors: Database

Sequence of Events:

1. User Selects to view Music and Lyrics
 - a. Systems presents a searchable interface with presets for genres, popular, liked songs, previous searches, etc.
2. User selects a song from the list provided (by search)
 - a. System displays the “song page” with a play button and lyrics at the bottom
3. User clicks the “Play” button
 - a. System processes the audio data and outputs to the speakers

Alternatives: The user could start the search over, repeating step 2 several times.

Exceptions: The user attempts to play a song that has missing audio data

Use case ID: 2

Name: Account Settings

Initiated by: User

Triggering Event: User Selection

Additional Actors: n/a

Sequence of Events:

1. User selects “Settings”
 - a. System presents the user with additional options including password management, personal information management, social media linking, audio settings, etc.
2. User selects a specific setting they want to manage
 - a. System retrieves the appropriate interface and displays
3. User edits the setting
 - a. System saves the selection and updates the user account

Alternatives: n/a

Exceptions: The system could fail to save the selected setting, resulting in the information the user set to be ‘forgotten’.

Use case ID: 3

Name: Room

Initiated by: User

Triggering Event: User selection

Additional Actors: Database, server, audience

Sequence of Events:

1. User Selects to join a “Room”
 - a. System connects the user to the server

- b. System displays the “Room” interface
- 2. User selects a song from the interface
 - a. System retrieves the song data from the database
 - b. System plays the song and displays the lyrics
- 3. User sings the lyrics
 - a. System records user as audio input
 - b. System sends the audio data to the server
- 4. Upon song completion, User selects to share their song
 - a. System sends the recorded data to the social media selected
 - b. System then displays the ‘audience’ room interface

Alternatives: The User may join the ‘audience’ upon selecting to join the room. They would then wait to start at step 2. The user may or may not share their song at step 4 and skip the first system response.

Exceptions: User fails to join a room, either by failing to secure a connection to the server or if the room is full.

Use case ID: 4
 Name: Save and Share
 Initiated by: User
 Triggering Event: User selection
 Additional Actors: database

- Sequence of Events:
- 1. User selects to “share”
 - a. System retrieves previously saved and shared data
 - 2. User selects to share saved data
 - a. System sends saved data to the linked social media
 - 3. User selects to delete saved data
 - a. System removes the saved data from the user’s device

Alternatives: The user may do either or both steps 2 and 3 and in a different order.

Exceptions: The saved data is corrupted upon retrieval and is unable to be shared or is shared improperly.

Use case ID: 5
 Name: Song Catalog
 Initiated by: Database
 Triggering Event: catalog maintenance
 Additional Actors: n/a

Sequence of Events:

1. Database prompts to add/remove a song
 - a. System allocates space and inserts the song into the current catalog
2. Database prompts to add the lyric data to the song data
 - a. System allocates space and inserts the lyrics with a link to the song

Alternatives: n/a

Exceptions: The system fails to properly link the lyrics to the song, resulting in a song with incorrect or no lyrics. The system fails to allocate enough space for the song/lyrics, resulting in incomplete data storage/corruption.

10 Functional Requirements

Requirement F1- Music and Lyrics search

Description:

Allow the user to search for songs and listen to them with lyrics displayed.

Rationale:

The song searched for needs to exist within the song catalog stored in the database.

Fit Criterion:

Search and retrieval will have reasonable access times. The user should not be waiting for a search result for more than a couple of seconds.

Acceptance Test 1:

Check whether the song exists in the database.

Requirement F2- Connection

Description:

Allow the user to find a “room” via server connection and communication.

Rationale:

The user should be able to be in a karaoke room with their friends/strangers.

Fit Criterion:

The system will require adequate internet connection to communicate with the server in sending and receiving data.

Acceptance Test 2:

Check if the connection being made is valid and secure.

Acceptance Test 3:

Check if the user has an adequate internet connection to the server requested.

Requirement F3- Song Catalog

Description:

The maintenance user should be able to add and remove songs/lyrics to the main song catalog in the database.

Rationale:

The song catalog should be able to grow and change as the need arises.

Fit Criterion:

The system should allocate space for the song/lyric data and insert into the sorted database such that access time is reasonable.

Acceptance Test 4:

Test the access time for storing and retrieving the songs. If the request goes over the allotted time, fail the test.

11 Data Requirements

D-1 – Username Requirements

Description: Usernames for accounts must be unique.

Rationale: Users should be able to definitively distinguish two users from each other in order to efficiently find the friends they want to sing Karaoke with.

Fit Criterion: All usernames in the database containing the usernames are unique in regards to one another.

Acceptance Tests: 5 - Username Uniqueness Test

D-2 – Email Requirements

Description: Emails for accounts must be valid.

Rationale: Without a valid email there is no sure method of communication to the user.

Fit Criterion: All emails in the database containing the emails are unique in regards to one another.

Acceptance Tests: 6 - Email Validity Test

12 Performance Requirements

12a Speed and Latency Requirements

SL-1 - Microphone Transmission Speed Requirements

Description: Microphone input must be transmitted very quickly.

Rationale: If microphone input is not transmitted quickly, it will be out of sync with the music and the karaoke will not work.

Fit Criterion: The transmission of microphone input from where a user begins talking to when another user hears them takes less than one millisecond.

Acceptance Tests: 7 - Mic Speed Test

SL-2 - Song Instrumental Transmission Speed Requirements

Description: Song instrumentals must be transmitted to users very quickly.

Rationale: If instrumentals are not transmitted quickly, they will be out of sync with the microphone input and the karaoke will not work.

Fit Criterion: The transmission of the instrumental from when the song begins to when the user hears the song takes less than one millisecond.

Acceptance Tests: 8 – Instrumental Speed Test

12b Precision or Accuracy Requirements

PA-1 - Microphone Transmission Accuracy Requirements

Description: Microphone input transmission must be accurate.

Rationale: If the microphone input loses too much quality over the transmission, it will be noticeable to the ear and the karaoke quality will go down.

Fit Criterion: The original microphone input audio and the microphone audio that another user hears after transmission must be at least 99% the same.

Acceptance Tests: 9 - Microphone Accuracy Test

PA-2 - Instrumental Transmission Accuracy Requirements

Description: Song instrumental transmission must be accurate.

Rationale: If the instrumental loses too much quality over the transmission, it will be noticeable to the ear and the karaoke quality will go down.

Fit Criterion: The original instrumental audio and the instrumental audio that a user hears after transmission must be at least 99% the same.

Acceptance Tests: 10 - Instrumental Accuracy Test

12c Capacity Requirements

C-1 - Room User Capacity Requirement

Description: Each Karaoke room should be able to handle fifty users at once.

Rationale: The largest types of karaoke environments we want to emulate, such as karaoke bars or karaoke parties, might have as many as around fifty people in attendance.

Fit Criterion: All speed and accuracy tests must pass in a room that has fifty users connected.

Acceptance Tests: 7, 8, 9, 10

C-2 - Number of Rooms Requirement

Description: The application should be able to handle 500,000 concurrent rooms.

Rationale: If the application is successful, it will have a lot of users. 500,000 rooms at 50 people per room gives us a maximum concurrent user capacity of 2.5 million users. If we cannot handle at least this many, the growth of the application may be stopped soon after finding success.

Fit Criterion: All speed and accuracy tests must pass when there are 500,000 rooms created that have 50 users each.

Acceptance Tests: 7, 8, 9, 10

13 Dependability Requirements

13a Reliability Requirements

R-1- Server Outages

Description: There should be no more than three server outages per month.

Rationale: Users should be able get together and have an amusing time doing karaoke, so keeping the server outages at a minimal number per month allows users to have an enjoyable experience without being concerned about any interruption.

Fit Criterion: Improvement in minimizing server outages is only possible after the system has been deployed. The maintenance team will collect data such as the number of outages for each month, and a couple of months after deployment evaluate if improvement needs to be made. We can upgrade our infrastructure to reduce server overload, thus minimizing server outages.

Acceptance Tests: 11

13b Availability Requirements

A-1 - Uptime

Description: Server should have uptime 97% of the time.

Rationale: It is highly desirable to have the system running most of the time, since as developers we want to remove inconvenience for the user caused by a time constraint.

Fit Criterion: Data will be maintained on server uptime through the service team and every few months evaluated to see if uptime is consistent with 97%. Data will be collected through timestamps on the servers. Necessary actions such as reducing server overload or upgrading hardware will be taken if required after evaluation.

Acceptance Tests: 12

13c Robustness or Fault-Tolerance Requirements

R/FT-1 - Robustness

Description: Under low internet speed that is unable to provide high quality streaming, users should still be able to hear and listen to the singer given that the speed is sufficient.

Rationale: The streaming should be optimized to suit the user's internet speed, meaning that if internet speed is low, then the system gives more priority to delivering consistent lesser quality streaming over high video quality streaming with delay.

Fit Criterion: Video Streaming can be tested at different levels of internet speeds and optimized at each level to suit user experience.

Acceptance Tests: 13

13d Safety-Critical Requirements

SC-1 - Safe Environment

Description: The product will make sure its servers are in a safe environment and there isn't electrical overload.

Rationale: In order to ensure that servers running for a long time do not cause any accidents and the environment is safe for teams to maintain the servers or upgrade them.

Fit Criterion: An electrical engineering team will be needed to make sure that the environment is safe. The electrical engineering should routinely check the servers and electrical components to make sure the environment is safe. If the

environment is deemed unsafe, then necessary adjustments must be made to comply to safety standards.

Acceptance Tests: 14

14 Maintainability and Supportability Requirements

14a Maintenance Requirements

Requirement MS1- Upkeep

Description:

The product will have weekly catalog updates to add new songs and remove any if necessary

Rationale:

The song catalog should be able to grow and change as the need arises.

Fit Criterion:

The system should allocate space for the song/lyric data and insert into the sorted database such that access time is reasonable.

Acceptance Test: Acceptance test ID:4

Requirement MS2- Bug fixes

Description:

The product will have monthly updates to address bugs and other user space issues as they arise, in order of importance.

Rationale:

In order to attract and retain customers, continuous improvement of the user experience is necessary.

Fit Criterion:

Bugs that are flagged should also have a rank of priority. The most important bugs will be addressed first.

Acceptance Test 15:

Check whether the bug patch breaks any other part of the product. If any part crashes or fails a test, fail this test and revert any changes.

14b Supportability Requirements

Requirement MS3- User Support

Description:

The product will provide the user with tooltips upon their first time using the product in each interface.

Rationale:

New users will not inherently know how to navigate the product and will require basic assistance.

Fit Criterion:

The tooltips should not be too invasive and should be disabled after first use.

Acceptance Test 16:

Check whether the tooltips pop-up appropriately and disable after first use. If they reappear or are the wrong tooltip for the functionality, fail the test.

14c Adaptability RequirementsRequirement MS4- Hardware**Description:**

The product will require multiple I/O devices.

Rationale:

If the user does not have the hardware, they will not be able to have functionality.

Fit Criterion:

The user will need a microphone for audio input and some form of speaker/headphones for audio output.

Acceptance Test 17:

Check whether audio I/O are available and if the product has the permissions to use them. If not, fail the test.

Requirement MS5- Software**Description:**

The product will require compatible software and operating systems.

Rationale:

As operating systems update, supports and features are added and dropped. The product will not be able to support all versions of all software.

Fit Criterion:

The user will need their device to be running a version of their OS that is compatible with the product.

Acceptance Test 18:

Check whether the product's software version is compatible with the OS version of the user. If it is not, fail the test.

14d Scalability or Extensibility RequirementsRequirement MS6- Growth**Description:**

As new customers download the product, the number of servers for "Rooms" (use-case ID: 3) will need to match.

Rationale:

The users will not want to wait for a server to become available, and instead will want immediate connection once other users are found.

Fit Criterion:

The product should have enough server space for all clients to be connected at peak hours.

Acceptance Test 19:

Check if the server space is adequate for the concurrent number of users. If at any given time the number of users exceeds what the servers can handle, fail the test.

14e Longevity Requirements**Requirement MS7- Lifetime**

The product will continuously be updated with current songs from pop culture as well as older songs yet to be added. As such, the product should not fade out of popularity. Therefore, it is expected that there will be support for this product for 10 years. As such, the estimated budget will be around 800-900k over this time period.

Acceptance Test 20:

Check the top song charts of the countries the product is released in. If the catalog does not contain a song on the charts, fail the test.

15 Security Requirements.**15a Access Requirements****A-1 - Safe Data Access**

Description: Only the user should be able to access its own sensitive data such as password or email. If the user forgets his or her password, they can request to change it based on the email they provided.

Rationale: As product creator, we want our product to be safe and secure for the users to use. In order to create that trust with users, we have to ensure that the user can access their account securely.

Fit Criterion: The system will establish that the user can access information about a particular user only if the username provided and the password provided match with the one in the database, this will grant the user access to the data associated with the username. The system will ensure that each username is unique.

Acceptance Tests: 21

15b Integrity Requirements

I-1 - Prevent Incorrect Data

Description: The product shall be protected from incorrect data being introduced in the system.

Rationale: If incorrect data is introduced, this can create a lot of complications for the user and can cause our product to lose the trust of the user.

Fit Criterion: Ensure that incorrect data is not introduced into the system by having checking conditions and constraints while inputting data. This will ensure that data is entered correctly and securely without malicious intentions. A testing team can inspect for loopholes that can be used to introduce false data, and fix those loopholes.

Acceptance Tests: 22

15c Privacy Requirements

P-1 - Information Practices

Description: The product shall make its users aware of its information practices before collecting data from them.

Rationale: It is important that users know what will happen to their own data, whether how the data is processed or where it is stored. Informing the user what will happen with their data will maintain the integrity of our organization.

Fit Criterion: When the user first creates their account, they should be informed about our organization's information practices. If changes occur to our information practices, the user should be notified via email.

Acceptance Tests: 23

15d Audit Requirements

Au-1 - Record User Actions

Description: The product may record user actions with their corresponding timestamps. This can include date the user created the account, the timestamps indicating the rooms they joined and the time they left the room.

Rationale: Records of user actions are kept to see how the users use the product and if any recent change in the product has increased user usage or decreased it.

Fit Criterion: The database should record user actions that can be later processed to gather data and make statistical inferences if the data is relevant. The developers can check that data regarding user actions is being stored.

Acceptance Tests: 24

15e Immunity Requirements

Im-1 - Safe Requests

Description: The product should make sure the connection between the user and server is secure and the data is encrypted. The server should make sure the requests of users are in accordance with expected requests.

Rationale: The product should be secure, and the user should feel safe using the product without having to worry about what might happen to their data. The system should be secure from various attacks to ensure the integrity of the product and organization.

Fit Criterion: The server should check that requests from users are verifiable and expected in order to make sure the requests are free from malicious intentions.

Acceptance Tests: 25

16 Usability and Humanity Requirements

16a Ease of Use Requirements

EU-1 - Song Searching Requirements

Description: Searching for the desired song instrumental should be simple and easy.

Rationale: Users may get frustrated and stop using the application if they can't find their song instrumentals easily.

Fit Criterion: The user's desired instrumental shows up in the top ten search results 90% of the time.

Acceptance Tests: 26 - Ease of Song Finding

16b Personalization and Internationalization Requirements

PI-1 - Settings Page

Description: There should be a single settings page with personalization settings such as language, profile picture editing, etc.

Rationale: A single settings page will be the simplest and easiest way for users to personalize the application.

Fit Criterion: There is a single settings page that has at least five personalization options.

Acceptance Tests: 27 - Settings Page Test

16c Learning Requirements

Learn-1 - Tutorial

Description: There should be an introductory tutorial to get users on the right page in regards to how to use the application.

Rationale: We want users to be able to use the application effectively very quickly, and an introductory tutorial is a good way to do that.

Fit Criterion: There exists an introductory tutorial after account creation that details how to use the application, with points such as how creating/joining a room works, and how to use the core important features in the application.

Acceptance Tests: 28 - Intro Tutorial Test

16d Understandability and Politeness Requirements

UA-1 - Readable Font Test

Description: The font is easily readable to users.

Rationale: If the font is unreadable, then the application is made unnecessarily hard to use and some users will not use it.

Fit Criterion: The font is readable to a majority of users.

Acceptance Tests: 29 - Readable Font

16e Accessibility Requirements

ACCESS-1 - Colorblind Option

Description: A colorblind option should be available to allow full accessibility of the application to colorblind users.

Rationale: Colorblind users will be just as interested in karaoke as anyone else, so we do not want colorblind inaccessibility to stop them from using this application.

Fit Criterion: With the colorblind setting activated, 99% of colorblind people should be able to use the application without issues relating to their colorblindness.

Acceptance Tests: 30 - Colorblind Test

16f User Documentation Requirements

UD-1 - Tutorial Documentation

Description: A tutorial is provided as documentation for users to understand the application.

Rationale: We want users to be able to understand the application so that they can use it effectively.

Fit Criterion: A tutorial exists that explains the application's functionality.

Acceptance Tests: 28 - Intro Tutorial Test

16g Training Requirements

Training-1 - Tutorial Training

Description: A tutorial should be provided to train users on the product.

Rationale: We want users to be able to use the product effectively.

Fit Criterion: A tutorial exists that trains the users on how to use the application's functionality.

Acceptance Tests: 28 - Intro Tutorial Test

17 Look and Feel Requirements

17a Appearance Requirements

App-1 - Professional Appearance

Description: The product should be neutral and professional in terms of color, fonts, logos, and icon. The product should look serious and well made, it should be creative but also measured. The product should not be unprofessionally colored, and colors should instill a sense of trust in the product.

Rationale: Having a professional and serious appearance that creates a sense of trust and visual appealing in the user is important. First impression is very important in using the product, this product intends to win over the user through an appealing and professional impression of the product.

Fit Criterion: Choose people who are not in connection with the people who designed the appearance, if most agree that appearance is professional and impressive, then requirement is met.

Acceptance Tests: 31

17b Style Requirements

Style-1 - Classic Style

Description: The product should look classical and professional such that it asserts reliability and quality in the minds of the viewer.

Rationale: Classical and Professional style of the product asserts the idea that this product is well made and reliable. Style is very important in the user's decision to use the product, and styles that are not unprofessional tend to create the impression that this product is not reliable or fake. There are many fake websites and apps today with unprofessional styling, so a professional style is very important in gaining the user's trust.

Fit Criterion: Ask a group of people, who have little bias towards our product, the style the product resembles. If most people say adjectives that are similar to classical and professional, then the requirement has been met.

Acceptance Tests: 32

18 Operational and Environmental Requirements

18a Expected Physical Environment Requirement - All Environments

Description: The product should be able to be played in any environment where wi-fi is available. However, the user would need to be able to access the internet through the device he wished to use the application on. However, the target environment for this device is actually one's home environment, or any other closed spaces where loud noises are acceptable.

Rationale: Since the application is one meant to connect with friends who are far away, it must use the internet as a means of communication. Therefore, anywhere internet connection is available, it stands to reason that the application would work.

Fit Criterion: The product is still usable in various environments.

Acceptable Test 33: Check if the application works in various environments with various conditions. If the application works in all environments, accept. Otherwise reject.

18b Requirements for Interfacing with Adjacent Systems Requirement - Different Devices

Description: The product shall be accessible by all those who use Windows 10.

The product shall be accessible by all those who use macOS Big Sur Version 11.2.1.

The product shall be accessible by all those who use Apple phone iOS 14.4.1.

The product shall be accessible by all those who use Android OS 11.

Rationale: Different people have different devices which utilize different OS. To make sure the product can be used by a wider range of users, it must be usable on different OS.

Fit Criterion: Different devices can utilize the same application.

Description: The product shall be usable on any updates that should take place on the mentioned OS.

Rationale: Updates to OS happen on a regular basis, and if we do not make sure our product runs on the newer versions as well it is sure to die out very quickly.

Fit Criterion: Make sure that product does not crash on the next release of OS.

Acceptance Test 34: Check if the application runs on the current version of OS of Windows, Big Sur, iOS, Android OS without crashing. If it does, accept. Otherwise reject.

18c Productization Requirements

Requirement - Product

Description: The product shall be of a size of 100MB and can be downloaded via each OS' unique distribution platforms like Apple store and Playstore.

Rationale: The maximum size for PlayStore is 100MB and it is greater for the Apple Store. In order to get the application to work for both, it must be 100MB.

Fit Criterion: Check if the application can be uploaded onto Playstore.

Description: For Playstore, the product shall be distributed as a .ABB file.

Rationale: Play Store requires an application to be in the form of a .ABB file or an .APK.

Fit Criterion: Check if the application can be uploaded onto Playstore.

Description: For Apple Store, the product shall be distributed as a .ipa file.

Rationale: Apple Store requires an application to be in the form of a .ipa file.

Fit Criterion: Check if the application can be uploaded onto Apple Store.

Description: The product shall be able to be installed by an untrained user without the need of prior instructions. Users need only to press the “install” button on their device to download the application.

Rationale: There are many people who are not that familiar with technology. We want even these people to be able to enjoy our product.

Fit Criterion: Have an easy “install” button that installs the application onto the device.

Acceptance Test 35: Check if the application can be installed without problems on both Playstore and Apple Store. If it does, accept. Otherwise reject.

18d Release Requirements

Requirement - New Release

Description: Every time an OS updates, developers will make sure the product is still functional. If the new OS interferes with the existing product, developers will release a modified version of the project which fixes the issue.

Rationale: We want our product to last for a long time, and in order for that to happen it must not crash at every release of a new OS.

Fit Criterion: The product does not crash and is usable after every OS version release.

Description: Along with that issue, there will be a weekly system maintenance rundown in order to fix any bugs or inconsistencies in the system. This will happen even in the case no issues are reported in a form of preventive maintenance.

Rationale: We want our product to last for a long time, and in order for that to happen it is prudent to fix any issues.

Fit Criterion: No bugs exist after every maintenance check.

Description: Developers will strive to develop and release new updates to the application in a quarterly, that is every 4 months. The new updates will run for the currently existing OS version.

Rationale: In order that our users do not get bored of the product, add new features to maintain their interest in the product.

Fit Criterion: Check if there are new elements to the product after each update of the product.

Acceptance Test 36: Create a report that details every update's release date and functionality. If time between release date is consistent and change in functionality is noted, accept. Otherwise reject.

19 Cultural and Political Requirements

19a Cultural Requirements

Requirement - Multiple Cultures

Description: The product will not be offensive to any nation, religion, ethnicity, race, etc.

Rationale: Everyone, no matter who they are, should be allowed to enjoy our product. We do not want to subtract from a user's enjoyability by accidentally offending the user.

Fit Criterion: Have a small survey regarding the matter.

Description: The product shall be available in multiple languages.

Rationale: There might be users from all over the world who use this application, and not everyone knows English. In order that these users get the best experience, it is best to have multiple languages.

Fit Criterion: Have a feature that allows users to choose language of their choice.

Acceptance Test 37: Check if there are languages provided for the user to use. If there are, then accept. Otherwise, reject.

19b Political Requirements

Requirement - Neutral Political Party

Description: The product will use only American based servers.

Rationale: Using American based servers means that in order to maintain these servers, valuable jobs for American's would be created. The product is being made in the US, therefore it is prudent that jobs are created in the US.

Fit Criterion: Have bases in the US where the Servers are stored.

Description: The product will be neutral and not display any political views of any party.

Rationale: There might be users with different types of political views who use this application. In order that these users get the best experience, it is best to be neutral so that users do not get offended.

Acceptance Test 38: Check if any symbols used, color scheme and design match that of any existing political party. If it does not, accept. Otherwise reject.

20 Legal Requirements

20a Compliance Requirements

Requirement - Privacy

Description: All personal information will be handled in accordance with data protection laws. No outside party will be able to listen into a conversation between two or more parties or view their screens without explicit permission from parties involved.

Rationale: Everybody values privacy. No one would appreciate their personal information being seen by others.

Fit Criterion: Test and make sure that personal information can't be seen or accessed by other users.

Acceptance Test 39: Check if the product violates any existing data protection laws. If it does, reject. Otherwise accept.

20b Standards Requirements

Description: The product will conform to generally accepted standards by today's media.

Rationale: New products like these release all the time. People do not have the time to learn to operate a completely different product when they can use others which are similar enough to other existing products.

Fit Criterion: Compare and contrast to other existing similar products.

Acceptance Test 40: Survey a large number of people and check if the application conforms to the general norm. If yes, accept. Otherwise reject.

21 Requirements Acceptance Tests

21a Requirements – Test Correspondence Summary

Test	Requirements																
	F-1	F-2	F-3	D-1	D-2	SL-1	SL-2	PA-1	PA-2	C-1	C-2	R-1	A-1	R/FT - 1	SC - 1	MS-1	MS-2
Test 1	X																
Test 2		X															
Test 3		X															
Test 4			X													X	
Test 5				X													
Test 6					X												
Test 7						X				X	X						
Test 8							X			X	X						
Test 9								X		X	X						
Test 10									X	X	X						
Test 11												X					
Test 12													X				
Test 13														X			
Test 14															X		
Test 15																X	
Test 16																	X
Test 17																	
Test 18																	X
Test 19																	X
Test 20																	
Test 21																	X

Table 1 - Requirements - Acceptance Tests Correspondence Part 1

Test	Requirements																				
	I-1	P-1	Au-1	Im-1	EU-1	PI-1	Learn-1	UA-1	ACCESS-1	UD-1	Training-1	App-1	Style - 1	ENV-1	Diff Dev-1	Prod-1	New Rel-1	Mult Cult-1	Political-1	Priv-1	Stad Req-1
Test 22	X																				
Test 23		X																			
Test 24			X																		
Test 25				X																	
Test 26					X																
Test 27						X															
Test 28							X														
Test 29								X		X	X										
Test 30									X												
Test 31												X									
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Test 33														X							
Test 34															X						
Test 35																X					
Test 36																	X				
Test 37																		X			
Test 38																			X		
Test 39																				X	
Test 40																					X

Table 2 - Requirements - Acceptance Tests Correspondence Part 2

21b Acceptance Test Descriptions

1 – Song Selection Test

Description: Search the song catalog and find the requested song with lyrics. If the song does not exist, fail the test. Otherwise, the test passes.

2 – Server connection Test

Description: Check whether the user connection to the server is valid and secure. If it is not, fail the test.

3 – Connectivity Test

Description: Check the user's internet connection. If the upload and download speeds are adequate, pass the test. If the user has frequent packet loss, high ping, or other general connectivity issues, fail the test.

4 – Song Retrieval Test

Description: Using an upper bound for time allotment, test whether song retrieval from the catalog abides by the upper bound. If the request times out, fail the test. Otherwise, pass the test. This includes telling the user the song does not exist if that's the case.

5 - Username Uniqueness Test

Description: Do a database search with each username in the database. If more than one matching result comes back for any search, the test is failed. Otherwise, the test passes.

6 - Email Validity Test

Description: Iterate through each email in the database, and check (via an API call or other method) whether the email is valid or not. If any email is not valid, the test fails. If all emails are valid, the test passes.

7 - Mic Speed Test

Description: Start a timer, send a microphone transmission from one machine, end the timer when the second machine receives the transmission. If the time recorded is more than one millisecond, the test fails. Otherwise, the test passes.

8 - Instrumental Speed Test

Description: Start a timer, begin the instrumental transmission, end the timer when a "user" machine receives the transmission. If the time recorded is more than one millisecond, the test fails. Otherwise, the test passes.

9 - Microphone Accuracy Test

Description: Transmit two minutes of non-trivial audio over the microphone transmission system. Compare the original inputted audio with the audio that another user hears after the original audio has gone through the transmission system. If the two pieces of audio are not at least 99% the same, the test fails. Otherwise, the test passes.

10 - Instrumental Accuracy Test

Description: Transmit two minutes of non-trivial audio over the instrumental transmission system. Compare the original inputted audio with the audio that another user hears after the original audio has gone through the transmission system. If the two pieces of audio are not at least 99% the same, the test fails. Otherwise, the test passes.

11 – Minimal Server Outage Test

Description: Check if product has experienced more than three server outages for the past month, if yes reject. If it did not experience more than three outages for the past month, then accept.

12 – Uptime Test

Description: Check if product uptime is 97% or more, if so accept. Reject, if product uptime is less than 97%.

13 – Optimized Video Streaming Test

Description: Check if product optimizes streaming at different levels of internet speed, if so accept. If the product does not optimize streaming when there is room for improvement, reject.

14 – Safe Environment Test

Description: Check if servers are in a safe environment through inspection by the electrical engineering team, if so accept. If servers are not in a safe environment, reject.

15 – Bug Fix Test

Description: Check whether the bug patch breaks any other part of the product. If any part crashes or fails a test, fail this test, and revert any changes.

16 – Tooltip Test

Description: Check whether the tooltips pop-up appropriately and disable after first use. If they reappear or are the wrong tooltip for the functionality, fail the test.

17 – I/O Test

Description: Check whether audio I/O are available and if the product has the permissions to use them. If not, fail the test.

18 – Version Test

Description: Check whether the product's software version is compatible with the OS version of the user. If it is not, fail the test.

19 – Server Space Test

Description: Check if the server space is adequate for the concurrent number of users. If at any given time the number of users exceeds what the servers can handle, fail the test

20 – Top Charts Test

Description: Check the top song charts of the countries the product is released in. If the catalog does not contain a song on the charts, fail the test.

21 – Safe Data Access Test

Description: Check if access to user data is allowed only if the given username matches with the one in the database and the given password matches the associated password with the username in the database. If it is, then accept or else reject.

22 – Prevent Incorrect Test

Description: Check if any loopholes can be found to introduce false data into the system if there are rejected. If the system has no loopholes, then accept.

23 – Information Practice Notification Test

Description: Check that users are notified of information practices when they first create their account and notified via email when information practices are modified. Accept, if users are notified both times else reject since they are not notified both times.

24 – Record User Actions Test

Description: Check that when a user performs certain actions the complete data of those actions is stored in the database, if so accept. If data is not being stored or data is partial, then reject.

25 Safe Requests Test

Description: Check that the server can make sure user requests are verifiable and expected, if the server can then accept. If the server cannot, then reject.

26 - Ease of Song Finding

Description: Measure how often users click on a top ten song search result versus how often they click on a result outside the top ten or a button called, “Song Not Found.” If 90% or more of clicks are on a top ten search result, the test passes. Otherwise, the test fails.

27 - Settings Page Test

Description: If there exists a setting page with five or more settings options, the test passes. Otherwise, the test fails.

28 - Intro Tutorial Test

Description: If there exists a tutorial after account creation that explains joining/creating a room, and the core features needed to begin singing karaoke, then the test passes. Otherwise, the test fails.

29 - Readable Font

Description: Put together a diverse focus group of 35 people and ask them to rate the readability of the font used in the application on a scale of 1-10. If 95% or more rate the font readability as a 9 or 10, then the test passes. Otherwise, the test fails.

30 - Colorblind Test

Description: Get the 1-10 accessibility rating of 100 colorblind people under the activated colorblind setting. If 99% rate the accessibility as a 9 or 10, the test passes. Otherwise, the test fails.

31 – Professional Appearance Test

Description: Interview people with little or no bias, if most of them agree that appearance is professional and impressive, accept. If they do not think that appearance creates that feeling, then reject.

32 – Classic Style Test

Description: Interview people who have no bias towards the product and ask them what style the product reflects. If most people give a description that is similar to classical and professional, then accept. Reject the test, otherwise.

33 – All Environments

Description: Test the product's functionality in all different sort of environments.

34 – Different Devices

Description: Test whether the product is functional on multiple devices.

35 – Production

Description: Check whether the product is of size 100MB, and that it can be uploaded onto Play Store and Apple Store

36 – New Release

Description: Test whether the product is still functional after updates on OS.

37 – Multiple Cultures

Description: Test whether people of all different cultures can enjoy the product.

38 – Neutral Political Party

Description: Check to make sure the product shows no bias towards any particular political party.

39 – Privacy

Description: Check to make sure that the data stored in the product is fully private.

40 – Standard

Description: The product conforms to standard norms established by previous online meeting platforms.

III Design

22 Design Goals

The video streaming should be high quality.

The video stream should be fast.

The system should be able to handle lot of clients.

User Interface should be of high quality and simplistic.

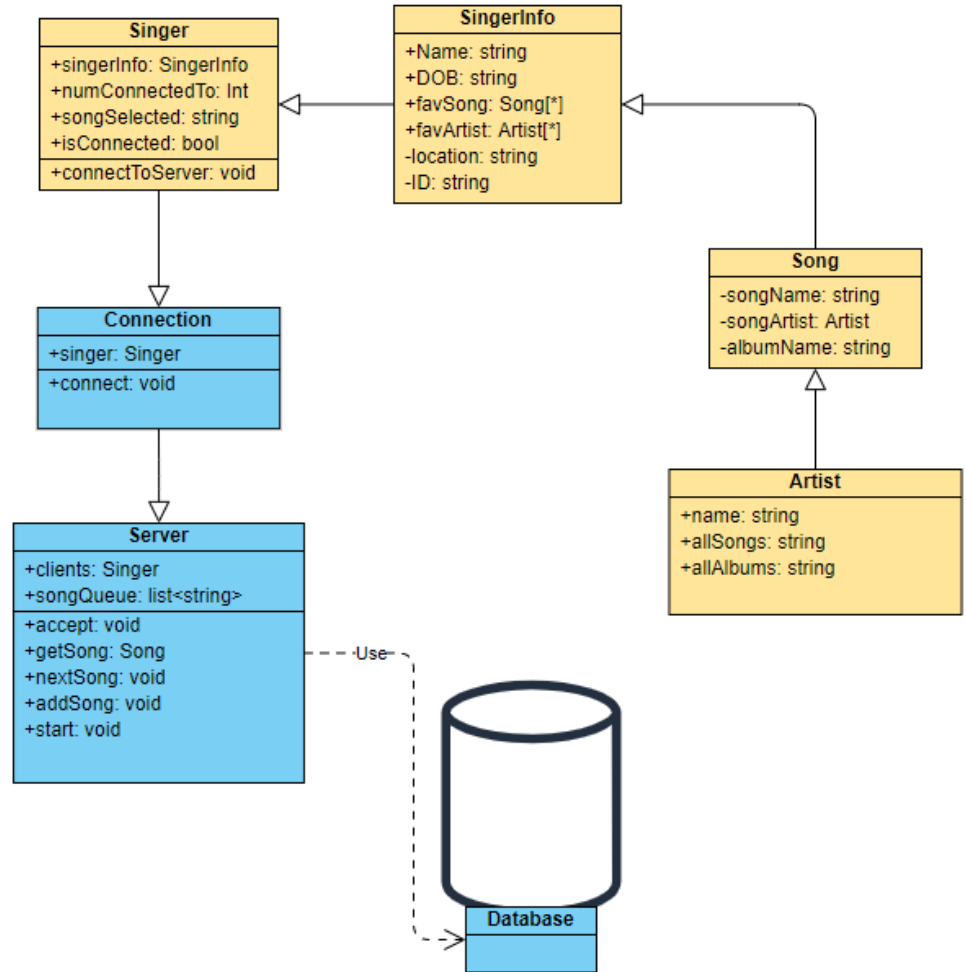
There will be more focus on making sure video streaming is high quality and it is fast, however since a very high-quality video could make the stream slower, our product must optimize video quality and speed of the stream in such a way that both goals are satisfied to the full extent possible.

23 Current System Design

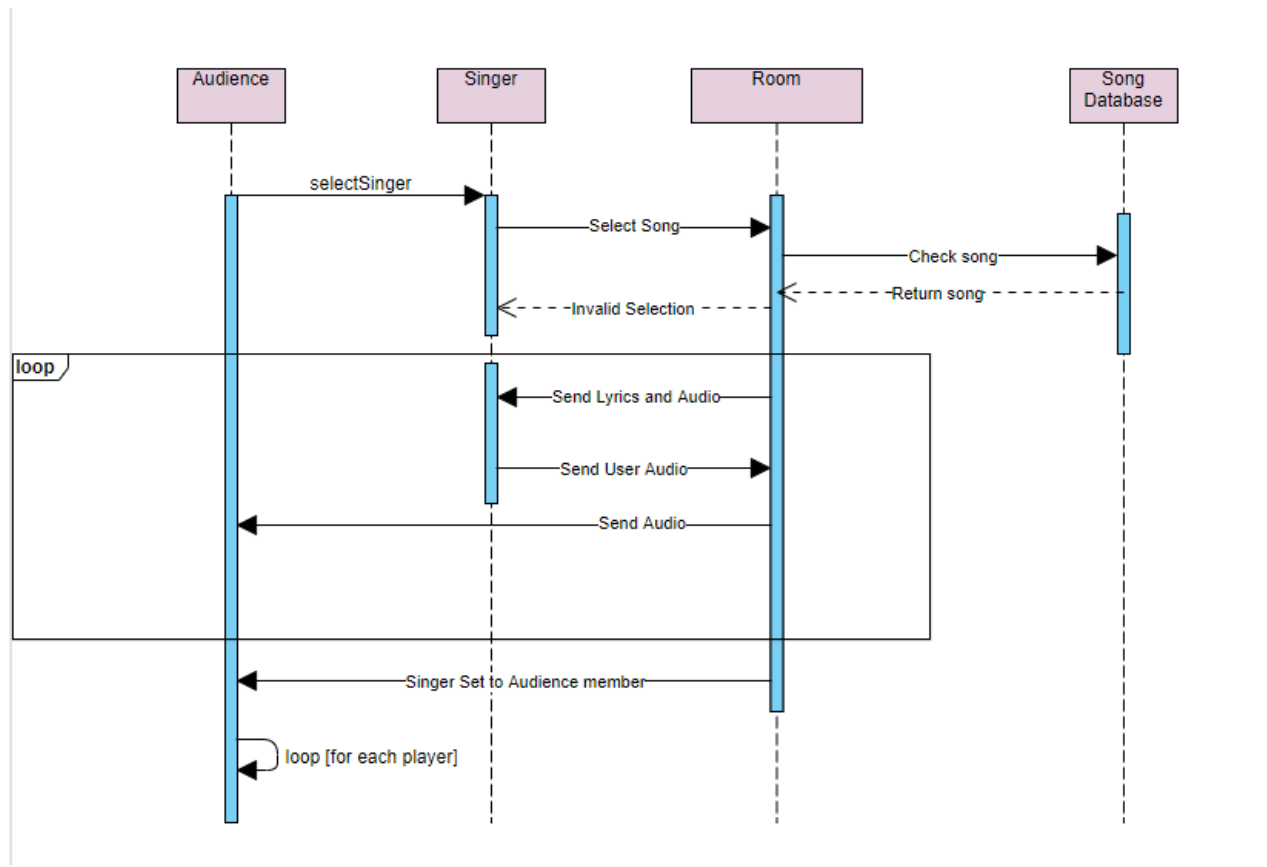
There is no pre-existing system.

24 Proposed System Design

24a Initial System Analysis and Class Identification



24b Dynamic Modelling of Use-Cases



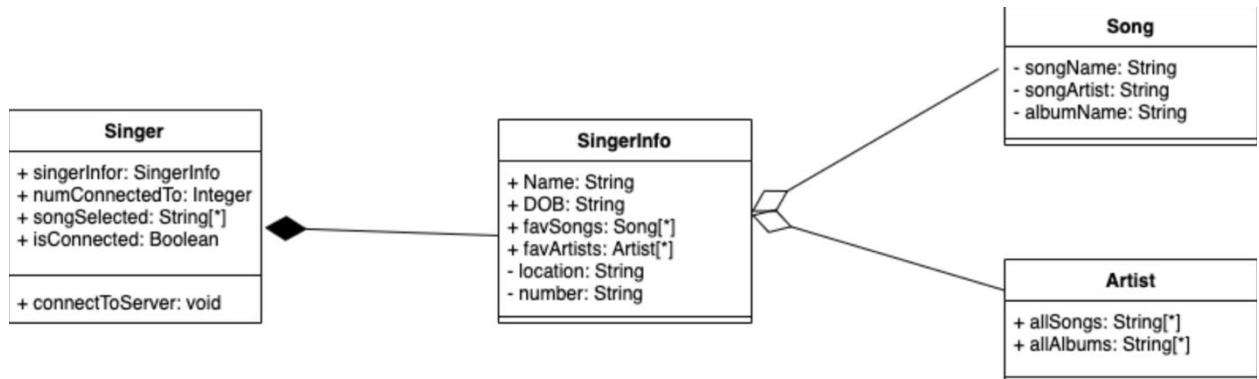
24c Proposed System Architecture

The basis of this entire project rests on it being a client-server model. The server in this case will be the “room” users connect to, and where they can do karaoke with each other. Each individual user is a client. There can be multiple rooms on a server, and there will be more than one server.

24d Initial Subsystem Decomposition

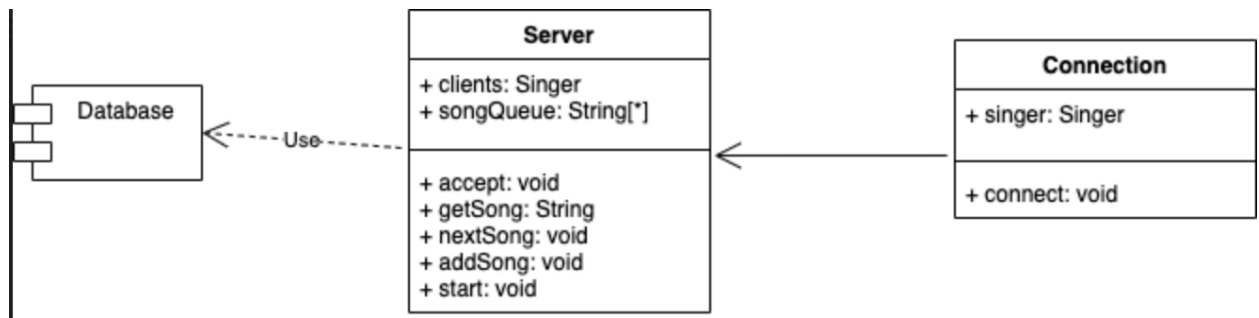
SINGER:

The Singer Subsystem represents how the user’s information is stored into the project.



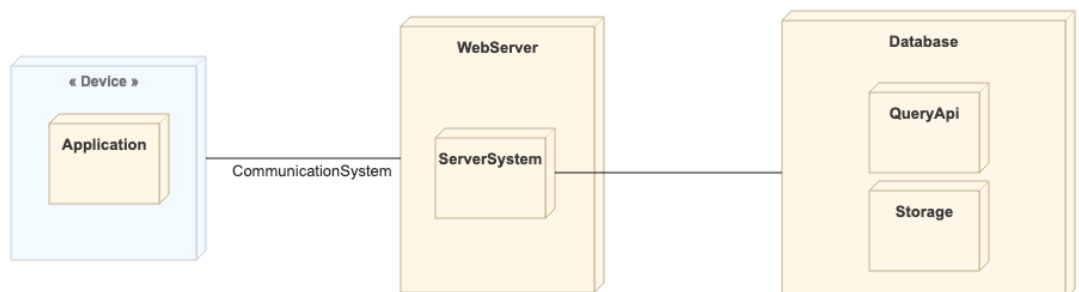
SERVER:

The Server Subsystem represents how a user connects to a karaoke room, and gets the services.



25 Additional Design Considerations

25a Hardware / Software Mapping



25b Persistent Data Management

A relational database will be used to store the data. In the database, there will be user data such as username, password, and email stored in tables. There will likely be also other data such as rooms that were created and users who were in the room, however the main priority will be given to user data. The user data can be accessed by the user after the user logs into the account, and the server will access obtain that data from the database. The server and the database will work together to make sure that data is stored safely, and it is also transported safely. The database will only response to query from the server, and the only way for the user to obtain data is through the services provided by the application and the server will respond to the requests made by the application.

25c Access Control and Security

Video streaming of the users that are in a room should be encrypted. Other type data shared among users across the internet must also be encrypted. Users can only access their own user data and that through only the server. Users can only interact with the server through the application API. The access control of users will be constrained through the application API. The Application API can send requests to the server that are valid and secure. Users can only see their own user data, and they can only use functionalities offered by the application API. In order to make application more secure, each request to the server must be verifiable meaning that the request server is being asked to perform is safe and something that a malicious attacker cannot exploit. Only the developers and the maintenance team can access the database and server. Servers and databases must be kept safe from attacks by malicious forces through limiting physical and virtual access to servers and databases.

25d Global Software Control

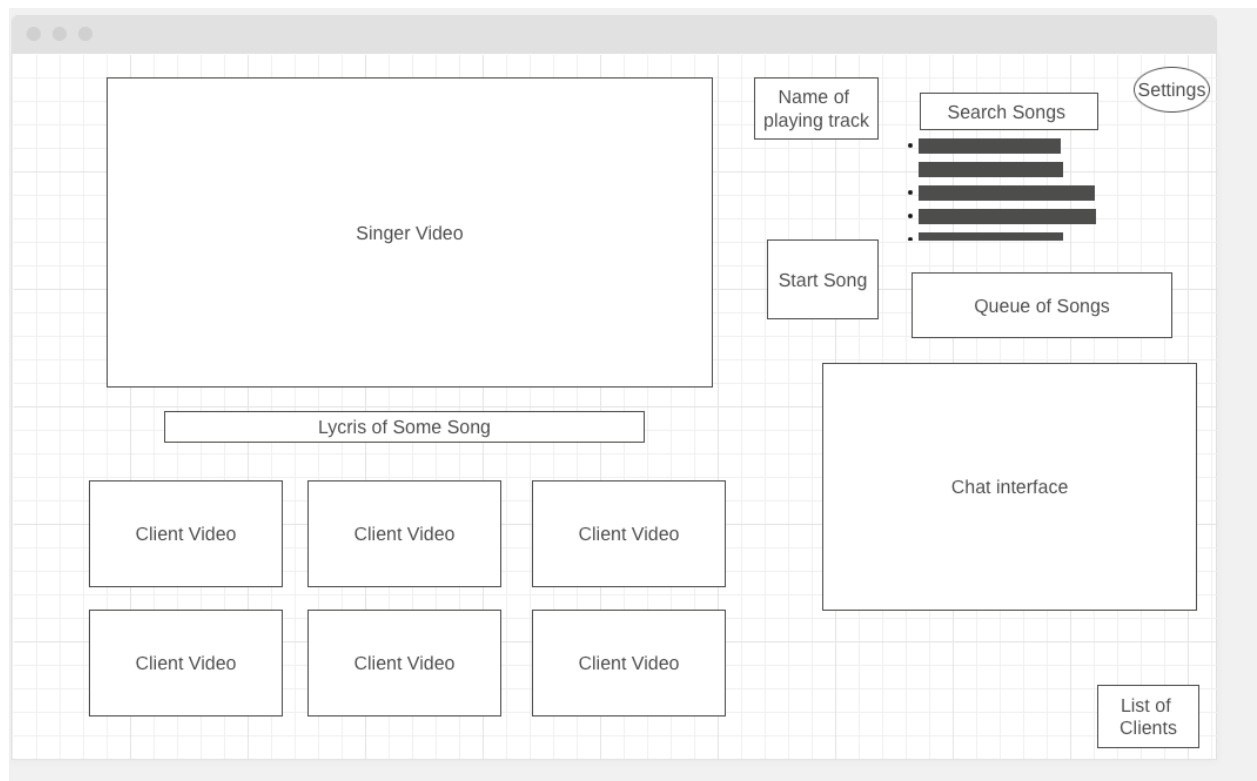
Since our project requires the use of many servers and databases due to capacity requirements, an additional subsystem is required to transfer the connection of a client from the central server to a worker server. Using a worker server will allow requests to be performed faster and decrease the workload on the central server. To organize many servers and databases, another subsystem must be created to simply organization and communication of servers and databases. Creating these

two subsystems will allow much more number of users to join rooms and stream video faster.

25e Boundary Conditions

If the system is unavailable because of maintenance or shutdowns, then the user must be notified when they are first opening the application. Important data such as username, email, and password must be safe from abrupt shutdowns, and in such situation user data will be the main priority.

25f User Interface



25g Application of Design Patterns

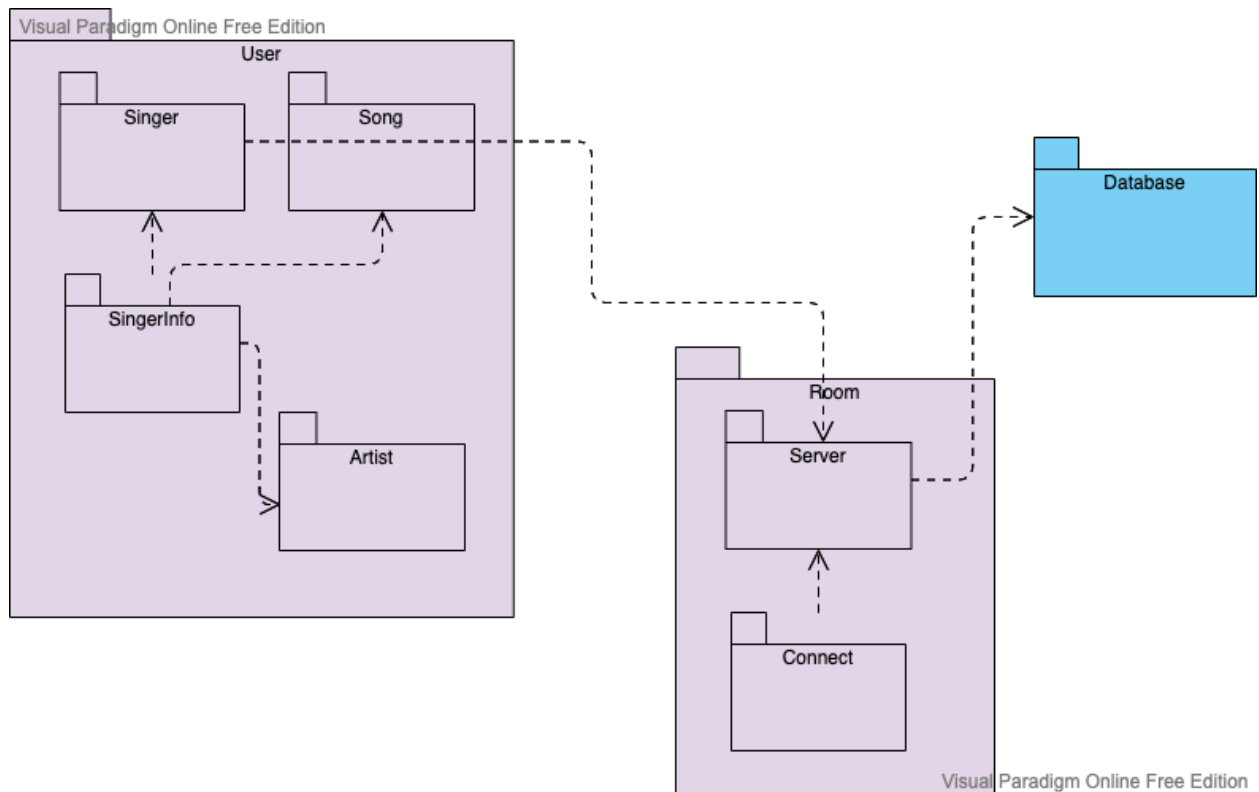
Since our project involves video streaming, a proxy design pattern will likely be used where a proxy object will contain a video. Playing a video is computationally expensive, so the proxy object will control the access to the video.

An observer design pattern will also be used since our clients will connect with the server in a “Room” when a particular event occurs in a room, we want all clients in the room to be notified. For example, if a user types a message in the chat, then all the other users will be notified since they are “observing” the chat.

Iterator design pattern will also be used because our product has a video streaming functionality each new video that goes to the client will be loaded into a linked list and play and this will happen constantly since video streams are usually very long. Also, when a user falls very behind in the stream due to low internet speed or other connections issues, we can clear the linked list, and wait for the new video stream to come.

Composite design pattern will also be used to design the user interface, since that is usually how Graphic User Interacts are designed. There will be a common superclass View, one subclass will be a concrete view, while other subclass will be a container view that can store many views. The leaves of the tree like hierarchy created will be concrete views.

26 Final System Design



27 Object Design

27a Packages

Two different packages can be implemented for this product. One package will hold classes which take in and store a user's account information. Another package holds classes related to a user connecting to an individual room, and the services the room provides.

27b Singer Subsystem

The singer subsystem is designed in order to store information regarding the user. There will be 4 classes implemented in this design. One class will hold information about the user, and what he is doing on the application at the moment. It will hold the user's personal information, whether he is connected to a room or not, a list of songs he plans on singing, and the number of people who are in a room with him. User's personal information will be stored and accessible through another class. This class will hold the user's name, date of birth, phone number, location, list of favorite songs and artists. There will be two more classes, one which holds song information like the name of the song, singer, and album, and another which holds artist information.

27c Room Subsystem

The room subsystem is designed in order to connect multiple users into a room and provide all the services a real karaoke room can. There will exist a class specifically designed to connect users to a room. It will have a Singer field, and a connect method to connect the users to a room. There will be a room/server class which can choose to accept the request for a connection or not. It will store a list of all users in the room, and queue of all the songs to be played. This room will

have the features of adding a song to the queue, removing a song, skipping songs, starting the music. In order to do these functions, the class will access a database which will store all information, including songs.

IV Project Issues

28 Open Issues

One issue is regarding what would happen to all the users when an external issue like network and servers going down. This will negatively affect user's experience as their time which should be spent singing with friends will be abruptly shut down.

Another issue is what would happen if someone does not follow proper protocols and shuts down their application while still in the room. Would their account simply leave the room, or will it cause problems to others in the same room.

29 Off-the-Shelf Solutions

29a Ready-Made Products

One ready-made product to use would be MySQL. It would be an off the shelf product used to manage a database.

Another ready-made product to use would be ManyCam. This would give the application webcam functionalities and background adjustment features for the user.

One invaluable ready-made product would be a microphone that can connect to the app. This would easily solve the issue on how to receive singing from the user.

29b Reusable Components

N/A

29c Products That Can Be Copied

While an application specifically for karaoke might be new, a virtual communication platform is not. There exist many applications like zoom or google hangouts which deal with connecting multiple people all over the world together. Products like these can be copied by grabbing software or ideas from these existing applications (legally of course).

30 New Problems

30a Effects on the Current Environment

The Current Environment for Karaoke usually happens at bars and restaurants where people get together and sing. Our product allows people to not meet physically and do Karaoke virtually through our product. Virtual karaoke could possibly replace physical karaoke given that everything is rapidly being digitized through software.

30b Effects on the Installed Systems

Considering that our product is designed for mobile devices such as android and iOS and Mac OS and Windows, there can be issues when communicating through many different platforms. Our product will ensure that it is compatible with the previously listed operating systems whether they are mobile, or laptop based.

30c Potential User Problems

No matter how we design our product, it is impossible to substitute the feeling that one gets doing karaoke physically. Regular karaoke performers will find it bit challenging to adjust to our version of virtual karaoke however overtime they will get use to our product. Users can encounter software issue that could occur since our product is cross-platform. Since our product requires high internet speed for users, when a high internet speed user is communicating with a low internet speed user then it can cause frustration if there is constant lagging and difficulty to communicate smoothly. Another user problem is that our product cannot provide all the songs on our platform due to copyright issues or other legalities that we will not be able to satisfy every user with our song options.

30d Limitations in the Anticipated Implementation Environment That May Inhibit the New Product

If too many people start using our product to a capacity, we are not prepared to handle due to budget constraints and lack of hardware infrastructure, then this cause limitations on the number of users that can use our product. Users with low internet will not be able to use this product to the full extent since our product is designed to suit high quality video streaming.

30e Follow-Up Problems

One of the main concerns is that how will our product provide high quality video stream since high quality video stream requires both a good internet speed and a good quality cam. Another issue could be the audio quality, could we provide high quality audio with the existing infrastructure of the user's device. Another issue is since some poor regions tend to have low internet speed would we make our product just Western-centric that could also cause other people to be upset with our product.

31 Migration to the New Product

31a Requirements for Migration to the New Product

If the application exceeds the expected server load, a migration to a new server system might be required. The old servers must continue to run in parallel as the new system is being implemented to retain current client connections. As the new servers come online, the old servers may be taken down in a 1 hour period that we will inform the clients about at least 1 week prior. During this 1 hour time window, all servers must be updated and brought back online.

31b Data That Has to Be Modified or Translated for the New System

The database for the product will contain all the application's songs and song information. A later version of the game may have a massive library to upkeep with millions of entries. If the current database no longer suits the needs of the application, migrating to a larger and/or more reliable database may be necessary. As such, all the old data must be backed up on a remote server to preserve all entries prior to the migration process. Losing any data may negatively impact the experience of the clients.

32 Risks

- Database requirements may exceed what is budgeted for
- Low initial sales may hinder the ability to staff for maintenance
- Low quality sound and video, depending on the servers and backend code

- Low productivity with adding music to the library
- Creeping user requirements may go unaccounted for or mispredicted.

33 Costs

This application is estimated to cost between \$200,000 and \$250,000 to produce. This is derived from the large database requirements, as well as the backend structure needed to support high quality video and audio streaming in each room instance. There are not any unique features that will take long to develop, so it should take approximately 1.5 to 2 years to develop, test, and release the base product.

34 Waiting Room

Integrated autotune as a feature would directly add to the karaoke experience. Given the key of the music, the autotune would work live, and make the experience smoother for the singers. If the singers are hesitant to sing karaoke because of performance anxiety, having autotune available could help relieve that anxiety. This feature could be integrated as a more “advanced” option, since the key of the music is required for it to work correctly. In further releases the autotune feature could be made to have an automatic option, in which a process is created that searches for or algorithmically computes the key of the chosen song.

Further future functionality could include instrument input support, such as input from MIDI keyboards and electric guitars. So, people could use the application to play along with songs on their instruments, or even duet with singers or other instrumentalists. Live-band karaoke could be a feature that comes from this.

35 Ideas for Solutions

Using a memory-efficient language like C or C++ could be a good decision, as lots of data will be moving in and out of memory; since our target audience are users that do not necessarily have powerful computers, we want to make sure the stress placed on their machine by the application does not cause them to cease further use. Beyond this, possible solutions vary widely and should be chosen based on the skills of the available developer pool.

36 Project Retrospective

When we started, our first order of business was to set up a Discord group to facilitate communication. This proved to be a good move, as from that point on we all had a constant channel of communication with each other. This channel of communication, along with weekly online meetings also via Discord, allowed us to always be on the same page with what we needed to accomplish and what we were each working on. Because of this effective communication, we were able to make consistent progress and end up with a document we are happy with.

However, we had one real stumble that fell within the realm of technical difficulty. The stumble was in regard to trying to combine separate pieces of this document into one. We tried to use GitHub for this purpose, but overwritten work and unfixable merge conflicts made our weekly meeting five and a half hours long, instead of normal one or two hours. After that meeting, we improved the process by simplifying it; we started sending the document around on Discord, and group members would add to the document one by one, in turns. This was an effective improvement. Beyond this, we had no serious difficulties.

V Glossary

Autotune: Software that changes the sung notes into the nearest correct notes. Correct notes are based on the provided musical key.

Database: An organized collection of data stored electronically.

Discord: Discord is a VoIP, instant messaging and digital distribution platform designed for creating communities.

Instrumental: The background music accompanied with a song without the lyrics without.

Karaoke: A type of interactive entertainment where people sing along to songs. A machine usually provides instrumental music along with lyrics.

MIDI: Musical Instrument Digital Interface; typically allows an instrument to interface with a computer.

Servers: A device that provides a functionality for other programs and devices.

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